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Passport: B01249874 **Gender:** Male **Date of birth:** 15/06/2000 **Nationality:** Nigeri an (Nigeria)

ABOUT ME

Environmental biologist and ecotoxicologist with a B.Sc. in Zoology and Environmental Biology, specializing in chemical risk assessment and environmental monitoring. Demonstrates expertise in toxicity testing (LC50 assays) from undergraduate research and hands-on laboratory toxicology from industrial training at the National Institute for Pharmaceutical Research and Development (NIPRD). Experienced in environmental administration at NCFRMI and public environmental health outreach through NYSC. Combines rigorous life-science research, community engagement, and full-stack software development to pursue postgraduate studies in environmental hazard management and global sustainability in Europe.

EDUCATION AND TRAIN-ING

[23/03/2019 – 14/03/2024]

Bachelor of Science in Zoology and Environmental Biology

University of Nigeria, Nsukka

City: Nsukka | **Country:** Nigeria | **Level in EQF:** EQF level 6

Core Academic Subjects:

- **Medical Entomology & Vector Control:** Surveillance. Monitoring and control mechanisms for biological vectors (e.g mosquitoes, flies) that transmit infectious diseases.
- **Parasitology & Tropical Diseases:** Life cycles. Host-parasite interactions, prevention strategies, and population control for endemic tropical diseases.
- **Epidemiological Methods & Quantitative Research:** Experimental research design. Biological sampling methodologies, biostatistics, and data interpretation for disease trends.
- **Environmental Health & Bio-Monitoring:** Assessing biological hazards, biohazard containment, toxicological assessment, and environmental exposure metrics affecting community health.
- **Biosafety & Laboratory Safety Protocols:** Decontamination procedures, chemical hazard management, safe specimen handling, and biological containment protocols.

Occupational skills:

- **Epidemiological Risk Assessment & Toxicity Testing:** Experienced in acute toxicity testing (LC50 determination), bioassay experimental setups, and water/ environmental quality analysis to evaluate public health hazards
- **Laboratory Operations & Biosafety Protocols:** Proficient in ethical specimen handling. Histology (microtome sectioning, H&E staining), cell motility analysis, biochemical screening assays (LFTs). Strict biohazard containment.
- **Community Health Education & Outreach:** Field experience in organizing public health awareness campaigns, educating communities in disease prevention, and coordinating volunteer teams during health drives.
- **Digital Health & Data Management:** Competent in using Python for quantitative data analysis, paired with full-stack software development (React, Node.js, Express.js, MongoDB9 to build platforms for resource routing and communication.

Principal Subjects:

- Public Health & Epidemiology: Applied parasitology & Public health (ZEB 421), General & Applied Entomology/vector Biology (ZEB 331).
- Environmental Health & Safety: Ecotoxicology & Pollution Biology (ZEB 452), Ecosystem & Waste Management (ZEB 454)

WORK EXPERIENCE

National Commission for Refugees, Migrants and Internally Displaced Persons (NCFRMI)

City: Abuja | **Country:** Nigeria

[06/01/2025 – 01/12/2025] **Community Service Staff (NYSC)**

- Welfare & Health Scheme Support: Assisted in delivering community welfare programs, health maintenance schemes, and basic healthcare access protocols for staff and displaced persons under commission care.
- Medical Logistics & Documentation: Processed medical requisitions, health insurance records, and humanitarian welfare assistance documentation to ensure timely relief and care delivery.
- Health Event Coordination: Coordinated logistics for field health drives, community awareness campaigns, and workplace wellness events targeting vulnerable groups and administrative teams. Administrative & Field Records Management: Managed daily departmental tracking logs, beneficiary health records, routine queries, and official inter-agency correspondence to maintain accurate program data.

National Institute for Pharmaceutical Research & Development (NIPRD) | Abuja, Nigeria

City: Abuja | **Country:** Nigeria

[05/2023 – 08/2023] **Industrial Research Trainee (Pharmacology & Toxicology)**

• **Animal House Operations & Handling:** Managed daily animal house operations, including ethical handling, feeding, housing maintenance, and routine health care for laboratory rodent models (mice and rats) used in experimental research.

• **Cell Motility & Microscopic Analysis:** Assessed cell motility and reproductive bio-functional parameters (including sperm motility evaluations) in mice and rats using compound and electron microscopy.

• **Detailed Histology & Organ Tissue Processing:** Executed end-to-end histology protocols: harvested target organ tissues, processed samples, performed microtome sectioning, and applied Hematoxylin & Eosin (H&E) staining to evaluate cellular tissue pathology and structural integrity.

• **Biochemical & Liver Toxicity Assays:** Conducted biochemical screening assays, including liver function tests (LFTs) for hepatotoxicity assessment and hepatitis-related biomarker evaluations.

• **Pharmacology & Pharmacokinetics:** Received structured practical training in fundamental pharmacology and pharmacokinetic concepts (absorption, distribution, metabolism, excretion - ADME), along with drug safety profiles.

• **Analytical Instrumentation:** Operated high-precision laboratory instruments, including High-Performance Liquid Chromatography (HPLC), UV-Vis Spectrophotometers, Particle Size Analyzers, and Dissolution Testers.

SKILLS

Microsoft Word | Microsoft Powerpoint | Microsoft Office | JavaScript Framework | Python (computer programming) | NoSql Databases (MongoDB) | HTML (Basics) | Express js | React js | assist employee health programmes | Toxicology assessment | histology | monitor water quality | health & safety | Version Control (Git)

LANGUAGE SKILLS

Mother tongue(s): Lingua igbo

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CERTIFICATIONS

[GOMYCODE, 15/06/2026] **Software Developer and AI skills**

Mode of learning: Project based

Link: https://diploma.gomycode.app/?id=31781560895333619

PROJECTS

[01/01/2024 – 05/05/2024] **Phytochemical Screening of Pterocarpus mildbraedii (Oha tree) & Acute Toxicity Testing on Juvenile Clarias gariepinus)**

- Research Focus: Evaluated the phytochemical composition of the aqueous stem bark extract of Pterocarpus mildbraedii (Oha tree) and its acute toxicity (96\text{-h } \text{LC}_{50}) on juvenile African catfish (Clarias gariepinus).
- Extraction Yield: The aqueous extraction yield of the powdered stem bark was 6.44%.
- Phytochemical Profile:
  - Qualitative screening confirmed the presence of alkaloids, flavonoids, saponins, terpenoids, steroids, phenols, phytates, tannins, glycosides, and carotenoids.
  - Quantitative analysis revealed alkaloids as the most abundant compound (16.10%), followed by flavonoids (1.60%), with lower concentrations of saponins (0.20%) down to trace amounts of carotenoids.
- Acute Toxicity Results:
  - No mortality occurred within the first 24 to 72 hours across all exposure concentrations (100–3000 mg/L).
  - At 96 hours, mortality was concentration-dependent: 0% mortality in 100–200 mg/L, 26.67% in 500 mg/L, 53.22% in 1500 mg/L, and 80% in 3000 mg/L.
  - The 96-hour median lethal concentration (\text{LC}_{50}) was calculated to be 2267.12 mg/L.
  - Estimated safe levels for aquatic ecosystems ranged between 0.02267 mg/L and 226.712 mg/L depending on the application factor applied.
- Observed Morphological & Behavioral Effects: Exposed fish displayed stress behaviors including erratic swimming, jerky movements, air gulping, loss of equilibrium, skin discoloration, and skin hemorrhage at higher concentrations (\ge 500\text{ mg/L}).
- Conclusion: The stem bark extract exhibits low acute toxicity (high \text{LC}_{50}), showing that toxic effects occur only at very high concentrations or after prolonged exposure.

Environmental Biology & Ecological Significance

- **Ecotoxicity & Non-Target Organism Protection:**
- **While** *Pterocarpus mildbraedii* ("Oha") contains bioactive secondary metabolites, evaluating its aquatic toxicity provides key ecotoxicological baseline data on how

botanical compounds affect non-target aquatic organisms. The high 96-h LC50 value (2,267.12 mg/L) confirms a low acute toxicity profile, indicating that typical environmental concentrations pose minimal immediate risk to freshwater fauna.

- **Aquatic Ecosystem Integrity & Habitat Quality:**
- **Runoff carrying concentrated botanical extracts or processing waste into lotic or lentic water bodies can alter water chemistry, elevate biological oxygen demand (BOD), and degrade critical aquatic habitats. Modeling maximum safe environmental concentrations (≤226.712 mg/L) helps prevent chemical stress in natural freshwater ecosystems.**
- **Bio-indicator Sensitivity & Trophic Dynamics:**
- *Clarias gariepinus* serves as a key sentinel species for aquatic ecosystem health and a vital component of Sub-Saharan freshwater trophic webs. Identifying behavioral and morphological endpoints—such as respiratory distress, epidermal hemorrhage, and loss of equilibrium at concentrations ≥500 mg/L—provides valuable biomarker data to detect early-stage physiological stress before population-level declines occur.
- **Ecotoxicological Baselines & Regulatory Environmental Standards:**
- **Establishing empirical safe discharge limits (0.02267 mg/L to 226.712 mg/L) via toxicological application factors provides actionable benchmarks for environmental monitoring, ecological risk assessment, and regional water quality management against botanical runoff.**
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[06/06/2026 – 07/10/2026] **Academiq.ng — Peer-to-Peer Academic & Tutoring Platform**

• Architected an end-to-end educational platform connecting students with academic tutors and study resources.

• Built real-time messaging, interactive tutor dashboards, and request management endpoints using Node.js, Express.js, MongoDB, React, and TypeScript.

• Implemented secure authentication routes, Redux state management, and responsive UI components.

Link: www.a-cade-miq.app

[16/01/2025 – 18/12/2025] **Breast Without Spots (BWS) Health Outreach & Cancer Awareness Campaign(Public Health)**

- Community-Based Educational Workshops: Interactive sessions organized with women's groups to build a network of "Breast Cancer Awareness" activists and teach proper techniques for self-breast examinations.
- Free and Low-Cost Health Screenings: Community clinical outreach programs conducted in partnership with local health organizations (such as the Alimosho community screening drive) to offer breast and cervical cancer screenings, consultations, and referrals.
- Policy and Advocacy Events: Focused meetings and roundtables engaging policymakers, tertiary healthcare providers, and diaspora organizations to advocate for sustainable medical resources, research, and skill transfers.
- Community Awareness Gatherings: Outreach events providing visual educational materials (charts, pamphlets, videos) and open forums where participants can share experiences and receive direct health education.